

# Practice Midterm

Phil 305 — Propositional and Modal Logic

## Part 1: S5

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Use the simplified rules for **S5** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S5**.

1.  $\Box(\Box p \rightarrow (q \vee r)) \vee \Box(\Box q \rightarrow p)$
2.  $\Diamond\Box(p \rightarrow q) \rightarrow (\Box p \rightarrow \Diamond q)$
3.  $\Box(p \vee q) \rightarrow (\Diamond(\neg q \vee r) \rightarrow \Diamond(p \vee r))$
4.  $(\neg\Diamond p \vee \neg\Diamond\Diamond p) \rightarrow \neg\Diamond\Diamond\Diamond p$

## Part 2: K

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Use the rules for **K** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **K**.

5.  $(\Diamond p \wedge \Box q) \rightarrow \Diamond q$
6.  $\Box(p \wedge \neg p) \vee \Diamond(q \vee \neg q)$
7.  $\Box\neg(p \vee \neg q) \rightarrow \Box p$

## Part 3: S4

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Use the rules for **S4** to build tableaux for each of these claims. Say which of them are valid, i.e., are theorems of **S4**. Remember **S4** is the logic **KT4**, so it has the rules for **T** and **4** as well as **K**.

8.  $\Box(p \rightarrow (q \wedge r)) \rightarrow (\Diamond\Diamond p \rightarrow \Diamond(q \vee s))$
9.  $(\Diamond\Box p \wedge \Diamond\Box q) \rightarrow \Diamond(p \wedge q)$
10.  $\Box\Diamond\Box\Diamond p \rightarrow \Box\Diamond p$